

SERTIS: QA Take Home Test

Duration : 7 days since the test was received

Feel free to use any tools, technologies, programming languages or libraries to complete the exercises below. In case that we don't provide enough information, you can make assumptions and keep a note about it in your answer.

We encourage you to give your best effort, and please know that it is fine if you are unable to complete every task. We are more interested in seeing how you approach the problem and the quality of your work than in complete solutions to every challenge.

If anything is unclear, feel free to reach out to us to clarify. Upon completion, please submit all relevant code and notes/documentation.

1. Imagine the team is tasked with testing the functionality of this [website](#)
 - a. Please list out the tasks required to test the website and estimate how long each should take (in days). In addition, please prioritize each task by importance.
 - b. Design the end-to-end test case or test scenario for the product purchasing flow, i.e. the process from selecting one or more products, adding them to the cart, to submitting an order. Be sure to cover both the happy and unhappy path.
 - c. Develop an automated test script for the test case/test scenario which you designed in 'b'
 - d. How could you performance test this website? Please explain. (There is no need to actually implement the performance tests, but feel free if you want to)
2. You'll be provided with a folder that contains a **local API server** and a **README** with setup instructions. Run this server locally, then freely design and implement your own API test against it.

The server exposes three endpoints:

- **[GET] /user/ids** - Get all user ids
- **[GET] /user/:id** - Get user information for the specified user id
- **[POST] /signin** - Sign in by specifying "phone_no" and "otp" of a particular user id in a request body

To try the API, you can first get a list of all user ids, to then get the information of a specific user, and use it to perform the sign-in. API Documentation can be found in the appendix below.

3. We would like to further automate the test(s) you created in exercise **1** and **2**. Please create a CI pipeline configuration which automatically runs the tests upon merging to a specific branch. We use Gitlab CI in our team, but you are free to

use any CI/CD tool you like (e.g. Gitlab CI, Github Actions, Jenkins, CircleCI, etc.)

Appendix: API Documentation

JavaScript

openapi: 3.0.3

info:

title: Sertis QA API

description: |-

This is a QA API Server for Sertis's QA Take Home Test

To try the API, you can first get a list of all user ids, to then get the information of a specific user, and use it to perform the sign-in.

Please create an automated test to test the functionality of the `"/signin"` endpoint.

contact:

email: nroon@sertiscorp.com

license:

name: Apache 2.0

url: <http://www.apache.org/licenses/LICENSE-2.0.html>

version: 1.0.0

externalDocs:

description: Find out more about Sertis

url: <https://www.sertiscorp.com/>

servers:

- url: <https://sertis-qa.glitch.me/>

tags:

- name: user

description: Operations about user

paths:

/user/ids:

get:

tags:

- user

summary: Get user ids

description: Get all available user ids in the system

responses:

'200':

description: A JSON object contains all available user ids

content:

application/json:

schema:

type: array

items:

type: string

example: ["01", "02", "03"]

'5XX':

description: Unexpected error

/user/{id}:

get:

tags:

- user

summary: Get a user by ID
description: Get all available user ids **in** the system
parameters:

- **in**: path
- name**: id
- schema:
 - type: string
 - required: true
 - description: ID of the user to get

responses:

'200':
description: A **JSON** object of a particular user
content:
application/json:
schema:

- type: object
- properties:
 - first_name:
 - type: string
 - last_name:
 - type: string
 - permission:
 - type: string
 - phone_no:
 - type: string
 - otp:
 - type: string

example:

- first_name: John
- last_name: Doe
- permission: admin
- phone_no: "**020011893**"
- otp: "**123456**"

'400':
description: Invalid user id
content:
application/json:
schema:

- type: object
- properties:
 - status_code:
 - type: string
 - message:
 - type: string

example:

- status_code: "**400**"
- message: "**123 is not a valid id**"

'5XX':
description: Unexpected error

/signin:
post:
tags:

- user

summary: Sign **in** user

```
description: Sign user into the system
requestBody:
  description: User object
  content:
    application/json:
      schema:
        type: object
        properties:
          phone_no:
            type: string
          otp:
            type: string
        example:
          phone_no: "020011893"
          otp: "012345"
responses:
  '200':
    description: successful operation
    content:
      application/json:
        schema:
          type: object
          properties:
            status_code:
              type: string
            status:
              type: string
            data:
              type: object
              properties:
                id:
                  type: string
                first_name:
                  type: string
                last_name:
                  type: string
                permission:
                  type: string
            message:
              type: string
          example:
            status_code: "200"
            status: "Pass"
            data:
              id: "01"
              first_name: QA
              last_name: Sertis
              permission: admin
            message: "Sign in success"
  '404':
    description: User not found
    content:
      application/json:
        schema:
          type: object
          properties:
```

```
status_code:
  type: string
status:
  type: string
data:
  type: object
message:
  type: string
example:
  status_code: "404"
  status: "Not found"
  data: {}
  message: "Sign in success"
'500':
description: Internal Server Error
content:
  application/json:
    schema:
      type: object
      properties:
        status_code:
          type: string
        status:
          type: string
        data:
          type: object
        message:
          type: string
      example:
        status_code: "500"
        status: "Fail"
        data: {}
        message: "Internal Server Error"
```